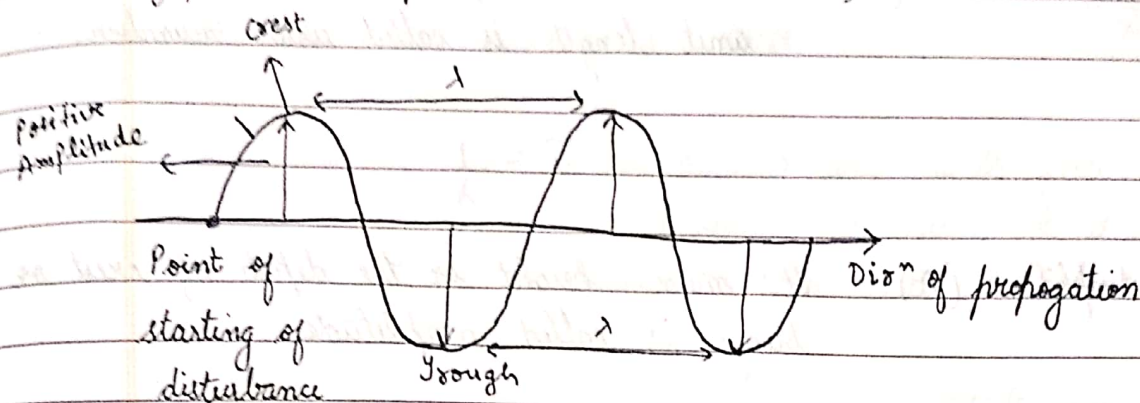


## Unit - 2 Oscillations and waves

Oscillatory Motion is the to and fro motion (Periodic disturbance of particles in space is called wave motion)



\* Wave motion  $\Rightarrow$  The particles periodic disturbance in the space is called wave motion and this motion represents the direction of propagation of the periodic disturbance which carries energy.

### \* Characteristics of waves

① Wavelength ( $\lambda$ ) = Distance (successive distance) b/w 2 crest or 2 trough from the reference line is called the  $\lambda$  of the wave and it is denoted by  $\lambda$ . Its unit can be m, mm, etc.

② Frequency ( $f$ ) = The total no of waves passing through the reference point per second is called the frequency of the wave. It is denoted by  $\nu$  and it is measured in Hz, cps.

$$1 \text{ KHz} = 10^3 \text{ Hz}, 10^3 \text{ cps}$$

$$1 \text{ MHz} = 10^6 \text{ Hz}, 10^6 \text{ cps}, \text{ s}^{-1}$$

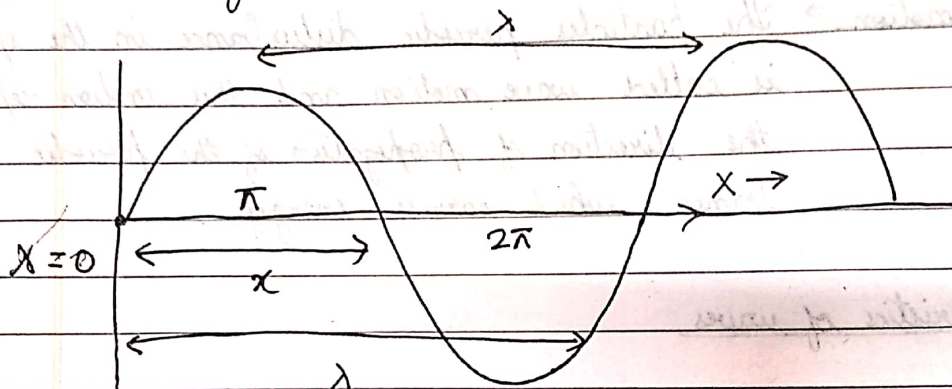
(3) Velocity ( $v$ ) = The total distance travelled by the wave per unit time is called velocity of the wave.

(4) Wave number ( $\bar{\nu}$ ) = The total no. of wave accommodated per unit length is called wave number.

$$\bar{\nu} = \frac{1}{\lambda}$$

(5) Amplitude ( $A$ ) = The max. height or the depth of crest or trough is called amplitude.

Wave Equation



$$\lambda \rightarrow 2\pi$$

$$x \rightarrow \phi$$

$$\lambda \phi = 2\pi x$$

$$\phi = \frac{2\pi x}{\lambda}$$

$$y = A \sin \left( \omega t + \frac{2\pi x}{\lambda} \right)$$

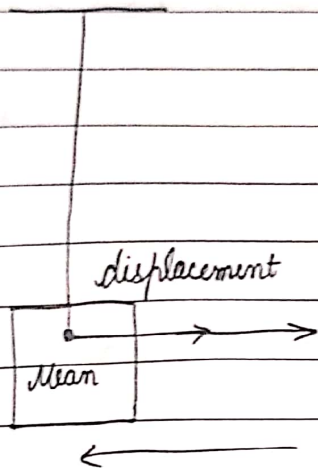
$$\omega = \frac{2\pi}{T}$$

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$$x = A \sin \left( \frac{2\pi}{T} t - \frac{2\pi}{\lambda} x \right)$$

$$x = A \sin 2\pi \left( \frac{t}{T} - \frac{x}{\lambda} \right)$$

Differential wave equation for SHM.



displacement force is the force which is bringing the spring to its mean position. That force is called restoring force.

Restoring  $\propto$  Displacement

$$m \cdot a \propto -x$$

$$m \frac{d^2x}{dt^2} \propto -x$$

$$m \frac{d^2x}{dt^2} = -kx$$

$$\boxed{\frac{d^2x}{dt^2} = -\frac{k}{m}x} \quad \text{Diff. form of SHM} \rightarrow (1)$$

$$\text{let } \frac{k}{m} = \omega^2 \Rightarrow \omega = \sqrt{\frac{k}{m}}$$

$$\boxed{\frac{d^2x}{dt^2} + \omega^2 x = 0} \rightarrow (2)$$

Now, the solution of (2) diff. eq<sup>n</sup> of SHM :-

$$\text{let } x = c e^{\alpha t} \text{ where } c \text{ and } \alpha \text{ are constants} \rightarrow (3)$$

$$\frac{dx}{dt} = c \alpha e^{\alpha t} \rightarrow (4)$$

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$$\frac{d^2x}{dt^2} = c d^2 e^{\alpha t} \rightarrow (5)$$

Using (3) & (5) in (2)

$$c d^2 e^{\alpha t} + w^2 c e^{\alpha t} = 0$$

$$c e^{\alpha t} (d^2 + w^2) = 0 \rightarrow (6)$$

$$c e^{\alpha t} \neq 0$$

$$\text{then, } d^2 + w^2 = 0$$

$$d^2 = -w^2$$

$$d = \sqrt{-w^2}$$

$$d = \pm w \sqrt{-1}$$

$$\boxed{d = \pm i w} \quad \text{Roots of } d$$

$$x = c e^{\alpha t}$$

$$= c e^{i w t} \text{ and } c e^{-i w t} \rightarrow (7)$$

\* If values of  $x$  from (7) satisfies (2) then linear comb<sup>n</sup> of (7) will also satisfies the equation (2).  
then,

$$x = c_1 e^{i w t} + c_2 e^{-i w t} \quad [\text{linear combination}] \rightarrow (8)$$

$$\text{we know } \Rightarrow \left. \begin{aligned} e^{i \theta} &= \cos \theta + i \sin \theta \\ e^{-i \theta} &= \cos \theta - i \sin \theta \end{aligned} \right\}$$

Put these in (8)

$$x = c_1 (\cos w t + i \sin w t) + c_2 (\cos w t - i \sin w t)$$

$$x = (c_1 + c_2) \cos w t + i (c_1 - c_2) \sin w t \rightarrow (9)$$

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$$\left. \begin{aligned} \text{Let } C_1 + C_2 &= A \sin \phi \\ \& i(C_1 - C_2) &= A \cos \phi \end{aligned} \right\} \rightarrow (10)$$

Put (10) in (9)

$$x = A \sin \phi \cos \omega t + A \cos \phi \sin \omega t$$

$$x \Rightarrow A \sin(\omega t + \phi) \quad \text{it is the solution of eq (2) (SHM wave differential eq)}$$

$$(1) \quad x_{\max} = A \times 1 \quad \left\{ \sin 90^\circ = 1 \right\} \quad \text{Maximum amplitude}$$

↓  
maximum

$$(2) \quad \text{Time period } (T = \frac{2\pi}{\omega})$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{k}{m}}} = 2\pi \sqrt{\frac{m}{k}}$$

$$(3) \quad \text{Frequency} = \frac{1}{T} \Rightarrow \frac{1}{2\pi \sqrt{\frac{m}{k}}}$$

$$(4) \quad \text{Wavelength } (\lambda) = \frac{c}{\nu} \Rightarrow \frac{c}{\frac{1}{2\pi \sqrt{\frac{m}{k}}}}$$

## Basic of Electromagnetic Theory

### \* Superposition principle

$$x_1 = A \sin(\omega_1 t + \phi_1)$$

$$x_2 = A \sin(\omega_2 t + \phi_2)$$

$$x_{\text{net}} = x_1 + x_2 + \dots$$

When 2 or more particles oscillates SHM then the net displacement is the sum of the displacement caused by individual displacement by ~~different~~ particles differently is called superposition of SHM.

## Basic of Electromagnetic theory

Electric + Magnetic  
Study of charge

### ① When charge is at rest

→ E · F

→ Coulomb's law

→ Gauss law

→ Electrostatics

### ② When charge is moving with const. velocity with $a = 0$

→ E · F

→ M · F

→ Current

③ When charge is moving with acc<sup>n</sup>,  $a \neq 0$

→ E·F

→ M·F

→ Radiation

\* what is field?

=> Invisible energy (3D spread of energy) in which a particle experience force is called field or the region or space in which particle experience force is called field.

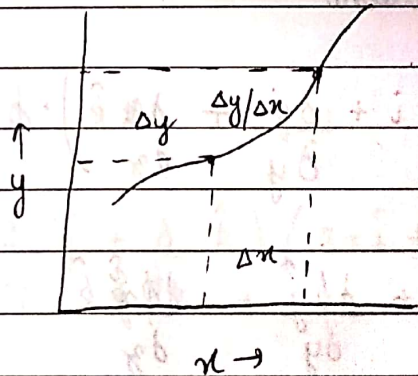
They are of 2 types :-

① Scalar field

② Vector field

Simple mathematical tools

① Derivative =  $\frac{\Delta y}{\Delta x}$



① Partial derivative =>  $\vec{\nabla}$  [Del operator]  
↳ Nebula

$$\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

Operator = Instruction

The rate of change of physical scalar quantity with distance is called gradient of a physical scalar quantity.

① When  $\vec{\nabla}$  is operated on a scalar field.

Let  $\phi = \phi[x, y, z]$  is a scalar quantity.

$$\vec{\nabla}\phi = \text{gradient of } \phi \text{ (grad-)}$$

② If  $\vec{A}$  is a vector quantity

$$\vec{\nabla} \cdot \vec{A} = \text{Divergence of } \vec{A}$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

③ For vector  $\vec{A}$

$$\vec{\nabla} \times \vec{A} = \text{curl of } \vec{A}$$

1) Gradient of Physical scalar quantity

$$\vec{\nabla}\phi = \text{gradient}$$

$$\vec{\nabla}\phi = \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot \phi(x, y, z)$$

$$\vec{\nabla}\phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$$

$$\phi(x, y, z)$$

$$d\phi = \left[ \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz \right]$$

$$= \left[ \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right] \cdot [dx \hat{i} + dy \hat{j} + dz \hat{k}]$$

$$d\phi = \vec{\nabla}\phi \cdot d\vec{r} \Rightarrow \boxed{|\vec{\nabla}\phi| |d\vec{r}| \cos \theta}$$

Q. Find  $\phi$  at  $\phi(x, y, z) = 3x^2 + 2y^3 + 4z^4$  at  $(1, 2, 3)$

$$\vec{\nabla} \phi = \left[ \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right] \cdot [3x^2 + 2y^3 + 4z^4]$$

$$\vec{\nabla} \phi = 6x + 6y^2 + 16z^3$$

$$= 6(1) + 6(2)^2 + 16(3)^3 \Rightarrow \boxed{462 = \vec{\nabla} \phi}$$

Divergence

$$\vec{\nabla} \cdot \vec{A} = \text{Divergence of } \vec{A}$$

$$\text{Divergence} = \frac{\text{Flux}}{\text{Volume}} = \text{Flux density}$$

Flux per volume is called divergence.

eg,  $\vec{A} = 3x\hat{i} + 4y\hat{j} + 3z\hat{k}$

$$\vec{\nabla} \cdot \vec{A} = ?$$

Ans,  $\vec{\nabla} \cdot \vec{A} = \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (3x\hat{i} + 4y\hat{j} + 3z\hat{k})$

$$\boxed{\vec{\nabla} \cdot \vec{A} = 10}$$

\*  $\vec{\nabla} \cdot \vec{A} = \text{positive}$ , then the system act as a source. The flux will come out from the source. [system is expanded]

\*  $\vec{\nabla} \cdot \vec{A} = \text{negative}$ , system for that act as a sink, negative flux, system is contracted.

$\vec{\nabla} \cdot \vec{A} = 0$ , then the system is incompressible, flux = 0, no expansion, no contraction.

\* Curl of a vector quantity

$$\Rightarrow \text{Curl } \vec{A} = \vec{\nabla} \times \vec{A}$$

\* Rotation / Circulation in a vector quantity is called curl of that vector quantity.

Calculation for the curl of the vector quantity.

$$\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

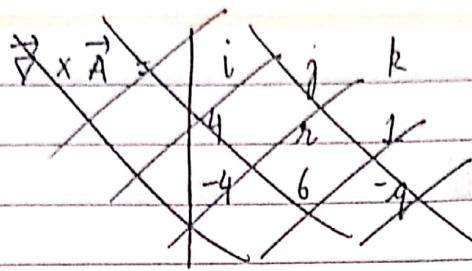
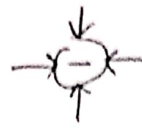
$$\text{Curl } \vec{A} \Rightarrow \hat{i} \left[ \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] - \hat{j} \left[ \frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right] + \hat{k} \left[ \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right]$$

Q1 Find the values of constants p, q and r if:-

$$\vec{A} = (4x - 4y + pz) \hat{i} + (rx + 6y - 2z) \hat{j} + (x - qy + 7z) \hat{k}$$

is an irrotational vector.

Ans. If  $\vec{A}$  is an irrotational vector, curl = 0.



$$\vec{\nabla} \times \vec{A} = \hat{i}[-q+2] + \hat{j}[x-p] + \hat{k}[r-4]$$

$$0 = \hat{i}[q+2] - \hat{j}[2-p] + \hat{k}[r-4]$$

$$\begin{aligned} -q+2 &= 0, & 1-p &= 0, & r+4 &= 0 \\ q &= -2, & 1 &= p, & r &= -4 \end{aligned}$$

\* Gauss Divergence theorem.

$$\iint_S \vec{A} \cdot d\vec{s} = \iiint_V (\vec{\nabla} \cdot \vec{A}) dV$$

→ It states that the line integral of a vector quantity over closed surface S equal to the volume integral of the divergence of the vector field or vector quantity over surface S enclosing volume V.

Proof →

$$\oint_S \vec{A} \cdot d\vec{s} = \int_{S_1} \vec{A} \cdot d\vec{s}_1 + \int_{S_2} \vec{A} \cdot d\vec{s}_2 + \dots + \int_{S_i} \vec{A} \cdot d\vec{s}_i$$

$$\oint_S \vec{A} \cdot d\vec{s} = \sum_{V_i \rightarrow 0} V_i \int_{S_i} \vec{A} \cdot d\vec{s} = \int_V (\vec{\nabla} \cdot \vec{A}) dV$$

Stokes law →

$$\oint_C \vec{A} \cdot d\vec{l} = \oint_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{s}$$

→ The line integral of a vector over a closed path is equal to line integral surface integral of curl of a vector over surface S.

Diff. form of Gauss law of electrostatics

$$\int_S \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0} = \frac{1}{\epsilon_0} \int_V \rho dv \left\{ \begin{array}{l} \text{the integrant must be vanish} \\ \text{as the integration is performing} \\ \text{arbitrary over the volume} \end{array} \right.$$

Using Gauss divergence theorem,

$$\int_V (\vec{\nabla} \cdot \vec{E}) dv = \frac{1}{\epsilon_0} \int_V \rho dv$$

$$\boxed{\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}}$$

$$\boxed{\vec{\nabla} \cdot \epsilon_0 \vec{E} = \rho}$$

$$\boxed{\vec{\nabla} \cdot \vec{D} = \rho} \quad \text{as } \epsilon_0 \vec{E} = \vec{D} \text{ (displacement vector)}$$

## # Gauss law of magnetostatics

$$\int_S \vec{B} \cdot d\vec{s} = 0$$

$$\int_V (\vec{\nabla} \cdot \vec{B}) dV = 0 \Rightarrow \boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

## # Equation of continuity

$$I = \frac{dq}{dt} \Rightarrow \text{If decrease in current then, } I = -\frac{dq}{dt} \rightarrow (1)$$

In terms of current density

$$\int_S \vec{J} \cdot d\vec{s} = -\frac{dq}{dt} \rightarrow (2)$$

$$\int_S \vec{J} \cdot d\vec{s} = -\frac{d}{dt} \int_V \rho dV \rightarrow (3)$$

Using Gauss divergence on LHS of eq (3)

$$\int_V [\vec{\nabla} \cdot \vec{J}] dV = -\frac{d}{dt} \int_V \rho dV$$

$$\int_V \left( \vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} \right) dV = 0$$

$$\boxed{\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0} \rightarrow (4) \text{ Varying Current}$$

H  $\Rightarrow$  Magnetic field Intensity  $\frac{1}{\mu_0}$

For steady state  $\Rightarrow \rho = \text{constant}$

$$\Rightarrow \frac{\partial \rho}{\partial t} = 0$$

$$\text{from (4)} \rightarrow [\nabla \cdot \vec{J} = 0]$$

Maxwell's Electromagnetic Equations.

- ①  $\nabla \cdot \vec{D} = \rho$  { Gauss law in electrostatics }
- ②  $\nabla \cdot \vec{B} = 0$  { Gauss law in magnetostatics }
- ③  $\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$  { Faraday law of Induction }
- ④  $\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$  { modify amp. circuital law }

Maxwell's equation in differential form.

Maxwell equations in integral form.

- ①  $\oint_S \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0} = \frac{1}{\epsilon_0} \int_V \rho \cdot dV \rightarrow ①$
- ②  $\oint_S \vec{B} \cdot d\vec{s} = 0 \rightarrow ②$
- ③  $\oint_L \vec{E} \cdot d\vec{l} = - \oint_S \vec{B} \cdot d\vec{s} \rightarrow ③$
- ④  $\oint_L \vec{H} \cdot d\vec{l} = \oint_S \left( \vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s} \rightarrow ④$

## # Physical significance of Maxwell's equations

Eq ①) Time independent equation.

- Divergence of  $\vec{D}$  is scalar. This implies  $\rho$  is also scalar.
- It relates space ~~static~~ variation of  $\vec{E}$  with  $\rho$ .
- It gives the statement of Gauss law of electrostatics.

Eq ②) Time independent equation

- It signifies that in an isolated magnet, monopoles do not exist.
- Number of magnetic field lines leaving and entering are equal.
- It gives statement of Gauss law of magnetism.

Eq ③) Time varying equation.

- It relates space variation of  $\vec{E}$  with  $\vec{B}$ . <sup>time variation of</sup>
- Time variation of  $\vec{B}$  generates  $\vec{E}$ .
- It gives the statement of Faraday law of induction and negative sign the justification of Lenz law.

Eq ④) Time dependent

- Magnetic field can be generated by vector  $\vec{J}$  and time variation of  $\vec{B}$  jointly or separately.
- It relates vector  $\vec{B}$  with  $\vec{J}$  and  $\vec{B}$ .
- It gives the statement of Ampere's law.

## # Derivation of Maxwell's 3rd equation.

To prove  $\Rightarrow \vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$

Prove, We know, that from FEMMI, emf will generate

$$e = -\frac{d\phi}{dt}$$

$$\oint_L \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{s}$$

Using Stokes theorem in LHS

$$\oint_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{s} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{s}$$

$$\boxed{\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}}$$

# Derivation of Maxwell's 4<sup>th</sup> equation.

To prove,  $\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$

prove,  $\oint_L \vec{B} \cdot d\vec{l} = \mu_0 I \rightarrow (1)$

$$\therefore H = B/\mu_0$$

$$\oint_L \vec{H} \cdot d\vec{l} = I \rightarrow \oint_S \vec{J} \cdot d\vec{s} \rightarrow (2)$$

By Stokes theorem.

$$\oint_S (\vec{\nabla} \times \vec{H}) \cdot d\vec{s} = \oint_S \vec{J} \cdot d\vec{s} \Rightarrow (3)$$

Div. of vector = 0

$$\vec{\nabla} \times \vec{H} = \vec{J} \rightarrow (4)$$

Taking divergence of (4) on both sides.

$$\underbrace{\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H})}_0 = \vec{\nabla} \cdot \vec{J} \rightarrow (5)$$

As from continuity equation

$$\vec{\nabla} \cdot \vec{J} = 0 \text{ means } \Rightarrow \vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

$$\frac{\partial \rho}{\partial t} = 0$$

Divergence of  $\vec{J} = 0$  which violates the continuity eq. and there is some need to modify the ampere's law.

then (4)  $\Rightarrow$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \vec{J}_d \rightarrow (7)$$

Now again taking Divergence on (7) both sides

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \vec{J}_d$$

$$0 = \vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \vec{J}_d$$

$$\vec{\nabla} \cdot \vec{J}_d = -\vec{\nabla} \cdot \vec{J}$$

$$= \frac{\partial \rho}{\partial t}$$

From Maxwell's 1<sup>st</sup> equation

$$\vec{\nabla} \cdot \vec{J}_d = \frac{d}{dt} (\vec{\nabla} \cdot \vec{D}) \Rightarrow \vec{\nabla} \cdot \frac{\partial \vec{D}}{\partial t}$$

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$$\text{Now } \vec{J}_D = \frac{\partial \vec{D}}{\partial t} \rightarrow (8)$$

Using (8) in (7)

$$\boxed{\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}} \quad \text{Modified Ampere Circuital law}$$

\* Work - Energy theorem OR Poynting theorem for E.M. waves

$$\begin{aligned} \text{Poynting vector } (\vec{S} \text{ or } \vec{P}) &= \frac{\text{Transportation of energy}}{\text{area} \times \text{time}} \\ &= W/m^2 \end{aligned}$$

$$\vec{S} = \vec{E} \times \vec{H} = \frac{\vec{E} \times \vec{B}}{\mu_0}$$

As we know, 4 maxwell equations can be written as:-

$$(1) \quad \vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$(2) \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$(3) \quad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu_0 \frac{\partial \vec{H}}{\partial t}$$

$$(4) \quad \vec{\nabla} \times \vec{H} = \vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

⇒ Taking the dot product (3) with H and (4) with E.

$$\vec{H} \cdot (\vec{\nabla} \times \vec{E}) = -\mu_0 \vec{H} \cdot \frac{\partial \vec{H}}{\partial t} \rightarrow (5)$$

$$\vec{E} \cdot (\vec{\nabla} \times \vec{H}) = \vec{J} \cdot \vec{E} + \epsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \rightarrow (6)$$

Now (4) - (2) (5) - (6)

$$\vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{H}) = -\mu_0 \vec{H} \cdot \frac{d\vec{H}}{dt} - \vec{J} \cdot \vec{E} - \epsilon_0 \vec{E} \cdot \frac{d\vec{E}}{dt}$$

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = -\epsilon_0 \vec{E} \cdot \frac{d\vec{E}}{dt} - \mu_0 \vec{H} \cdot \frac{d\vec{H}}{dt} - \vec{J} \cdot \vec{E} \rightarrow (7)$$

$$\left[ \text{energy in electric field} \right] \left[ \text{energy in magnetic field} \right]$$

$$\vec{E}^2 / 2 \quad E^2 = E \cdot E$$

$$\begin{aligned} \frac{\partial E^2}{\partial t} &= E \frac{dE}{dt} + \frac{dE}{dt} \cdot E \\ &= 2E \frac{dE}{dt} \end{aligned}$$

$$E \frac{dE}{dt} = \frac{1}{2} \frac{\partial E^2}{\partial t}$$

$$\text{Now, } \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = -\epsilon_0 \frac{\partial E^2}{\partial t}$$

$$= -\frac{d}{dt} \left[ \frac{1}{2} \epsilon_0 E^2 \right] - \frac{d}{dt} \left[ \frac{1}{2} \mu_0 H^2 \right] - \vec{J} \cdot \vec{E} \rightarrow (8)$$

Taking volume integration on equation 8 on both sides.

$$\int_V \vec{\nabla} \cdot (\vec{E} \times \vec{H}) dV = -\frac{d}{dt} \int_V \left\{ \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2} \mu_0 H^2 \right\} dV - \int_V (\vec{J} \cdot \vec{E}) dV$$

$\underbrace{\hspace{10em}}_{U_{\text{total}}} \quad \downarrow \textcircled{9}$

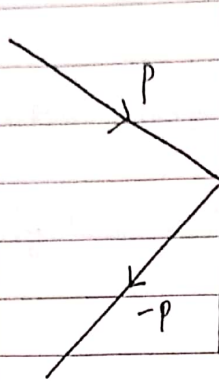
Total energy / flow ~~flow~~ leaving the volume  $V =$

Rate of decrease of energy stored in magnetic, electric field  
- ohmic power dissipated (work done).

$$U_{\text{total}} = U_e + U_m$$

$$\int_V (\vec{\nabla} \cdot \vec{S}) dV = - \int \frac{\partial U}{\partial t} \cdot dV - \int \vec{w} \cdot dV \rightarrow (10)$$

Momentum with EM waves.



$$\begin{aligned} E &= mc^2 \\ &= mc \cdot c \\ &= pc \end{aligned}$$

$$p = \frac{E}{c} = \frac{U}{c}, \quad p = \frac{-2U}{c} = \frac{2U}{c}$$

$$p = \epsilon_0 \mu_0 S$$

$$= \epsilon_0 \mu_0 \left\{ \frac{\vec{E} \times \vec{B}}{\mu_0} \right\}$$

$$\vec{p} = \epsilon_0 \{ \vec{E} \times \vec{B} \}$$

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times \{ \epsilon_0 (\vec{E} \times \vec{B}) \}$$

Q. Show that energy stored in M.F/unit volume in free space is  $\frac{B^2}{2\mu_0}$

Q. Assuming that energy from 1000 bulbs is radiated uniformly. Calculate the average intensity of E.F, M.F at a distance of 2m from the bulb.

Propagation of EM waves in free space

⇒ Plane wave is defined as whose amplitude is same at any point in the plane  $\perp$  to the specified direction.

- ① Calculate the velocity of EM waves in free space.
- ② To find the plane wave solution
- ③ To find the transverse nature of EM waves.
- ④ Phase of  $\vec{E}$  and  $\vec{H}$ .

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⑤ Impedance of EM waves.

⑥ Poynting vector

⑦ Energy vector flow.

(\*) As we know, 4 Maxwell's equations:-

$$① \quad \vec{\nabla} \cdot \vec{E} = \rho/\epsilon_0 = 0, \quad \rho = 0$$

$$② \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$③ \quad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu_0 \frac{d\vec{H}}{dt}$$

$$④ \quad \vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} = 0 + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \left\{ \vec{J} = 0 \right\}$$

Taking curl of ③ and ④

$$(\vec{\nabla} \times \vec{\nabla} \times \vec{E}) = -\mu_0 \frac{d}{dt} (\vec{\nabla} \times \vec{H})$$

$$\vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{d^2 \vec{E}}{dt^2} \quad \left| \because \vec{\nabla} \times \vec{H} = \epsilon_0 \frac{d\vec{E}}{dt} \right.$$

$$0 - \nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{d^2 \vec{E}}{dt^2}$$

or

$$\nabla^2 \vec{E} - \mu_0 \epsilon_0 \frac{d^2 \vec{E}}{dt^2} = 0 \rightarrow ⑤$$

Similarly for ④

$$\nabla^2 \vec{H} - \mu_0 \epsilon_0 \frac{d^2 \vec{H}}{dt^2} = 0 \rightarrow ⑥$$

⑤ & ⑥ are 2<sup>nd</sup> order differential equations representing eq<sup>n</sup> of a general wave.

$$\left[ \nabla^2 \psi - \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} = 0 \right] \rightarrow (7)$$

$$\text{As } v^2 = \frac{1}{\mu_0 \epsilon_0} \quad \therefore v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s (approx)}$$

then  $\Rightarrow v = c$  [speed of light]

from the above result, we can say em waves can travel with the speed of light. (5) & (6) becomes

$$\nabla^2 E - \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2} \rightarrow (8)$$

$$\nabla^2 H - \frac{1}{c^2} \frac{\partial^2 H}{\partial t^2} \rightarrow (9)$$

Now (8) and (9) of  $\vec{E}$  &  $\vec{H}$  can be expressed as :-

$$\vec{E}(x, t) = E_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \rightarrow (10)$$

$$\vec{H}(x, t) = H_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \rightarrow (11)$$

#  $E_0$  &  $H_0$  are constant amplitudes.

# where  $\vec{r}$  is the position vector  $\Rightarrow x\hat{i} + y\hat{j} + z\hat{k}$ .

# where  $\vec{k}$  is the propagation/wave vector  $\Rightarrow \frac{2\pi}{\lambda} \hat{n}$ .

$$\# \quad v = \frac{c}{\lambda} \Rightarrow \lambda = \frac{c}{v} \Rightarrow \frac{2\pi}{\lambda} = \frac{2\pi v}{c} \hat{n}$$

$$\boxed{k = \frac{\omega}{c} \hat{n}}$$

# To prove transverse nature, we will calculate divergence of  $\vec{E}$  &  $\vec{H} = 0$ .

$$\vec{\nabla} \cdot \vec{E} = 0$$

$$\vec{\nabla} \cdot \vec{H} = 0$$

$$\left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot \left[ E_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right] = 0$$

$$\hat{i} \left( E_0 x + E_0 y + E_0 z \right) \cdot e^{i(k \cdot r - \omega t)} \cdot [k_x x + k_y y + k_z z]$$

$$i \vec{k} \cdot \vec{E} = 0$$

$$\boxed{\vec{k} \cdot \vec{E} = 0}, \quad \boxed{\vec{k} \cdot \vec{H} = 0}$$

From the above result, both  $\vec{E}$  &  $\vec{H}$  are  $\perp$  to  $\vec{k}$ . This proves the transverse nature of em waves.

# To calculate phase of  $\vec{E}$  &  $\vec{H}$  we will find the curl of equation (10) and (11)

$$\nabla \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t}$$

we know,

$$\vec{H} = H_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\frac{\partial \vec{H}}{\partial t} = -i\omega H_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\frac{\partial \vec{H}}{\partial t} = -i\omega \vec{H}$$

$$\nabla \times \vec{E} = i\vec{k} \times \vec{E} = i\mu_0 \omega \vec{H}$$

$$\boxed{\vec{k} \times \vec{E} = \mu_0 \omega \vec{H}} \rightarrow (12)$$

$$\boxed{\vec{k} \times \vec{H} = -\epsilon_0 \omega \vec{E}} \rightarrow (13)$$

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Impedance ( $z_0$ )

⇒ From (12) we have

$$\vec{H} = \frac{\vec{R} \times \vec{E}}{\mu_0 \omega} = \frac{\hat{n} \times \vec{E}}{\mu_0 c} \quad \left| \quad \therefore \vec{R} = \frac{\omega}{c} \hat{n} \right.$$

Taking the modulus

$$|H| = \frac{E}{\mu_0 c} \rightarrow (14)$$

$$z_0 (\text{Impedance}) = \left| \frac{E}{H} \right| = \left| \frac{E_0}{H_0} \right|$$

$$\Rightarrow \mu_0 c = z_0 = \sqrt{\frac{\mu_0}{\epsilon_0}}$$

# Impedance in free space approx  $\approx 377 \text{ V/A}$

From the above result, in free space,  $\vec{E}$  &  $\vec{H}$  are in same phase and an impedance of  $377 \text{ V/A}$  will be offered by the given space to the EM waves.

# Poynting vector

$$\vec{S} = \vec{E} \times \vec{H} = \frac{\vec{E} \times \hat{n} \times \vec{E}}{\mu_0 c}$$

$$\Rightarrow \frac{\vec{E} \times \hat{n} \times \vec{E}}{z_0} = \frac{(E \cdot E) \hat{n} - (\hat{n} \cdot E) \vec{E}}{z_0}$$

$$\vec{S} \Rightarrow \frac{E^2 \hat{n}}{z_0} \Rightarrow \frac{E_0^2 \cos^2(k \cdot r - \omega t)}{z_0} \hat{n}$$

## Unit-1 & Unit-2 [Quiz]

$$\vec{S} = \frac{\vec{E}_0^2}{z_0} \cos^2(k \cdot \vec{r} - \omega t)$$

$$\langle \vec{S} \rangle = \left\langle \frac{E_0^2}{z_0} \right\rangle \left( \frac{1}{2} \right) = \frac{1}{2} \frac{E_0^2}{z_0} \quad \left| \because \langle \cos^2(k \cdot \vec{r} - \omega t) \rangle = \frac{1}{2} \right|$$

$$\Rightarrow \frac{E_{rms}^2}{z_0} = \langle \vec{S} \rangle \quad \left| \because E_{rms} = \frac{E_0}{\sqrt{2}} \right|$$

### # Energy flow

The energy density of a given space can be written as:-

$$\frac{u_e}{u_m} = \frac{\frac{1}{2} \epsilon_0 E^2}{\frac{1}{2} \mu_0 H^2} = \frac{\epsilon_0}{\mu_0} \times \frac{\mu_0}{\epsilon_0} = 1$$

$$u_t = u_e + u_m$$

if  $u_e = u_m$ , then,

$$u_t = 2u_e \Rightarrow \left( \frac{1}{2} \epsilon_0 E^2 \right) 2 = \epsilon_0 E^2 = u_t$$

$$\begin{aligned} \langle u_t \rangle &= \langle \epsilon_0 E^2 \rangle \\ &= \langle \epsilon_0 [E_0 \cos(k \cdot \vec{r} - \omega t)]^2 \rangle \\ &\Rightarrow \langle \epsilon_0 E_0^2 \cos^2(k \cdot \vec{r} - \omega t) \rangle \end{aligned}$$

$$\langle u_t \rangle \Rightarrow \epsilon_0^2 E_0 \frac{1}{2}$$

$$\langle u_t \rangle \Rightarrow \epsilon_0 E_{rms}^2$$

$$\begin{aligned} \text{Ratio of } \frac{\langle u \rangle}{\langle S \rangle} &= \frac{\epsilon_0}{z_0} = \frac{\epsilon_0}{\mu_0 c} \\ &= \epsilon_0 z_0 \end{aligned}$$

Q. Assuming that all the energy from 1000 V bulb is radiated uniformly. Calculate the average values of intensity of electric & magnetic field of radiations at a distance of 2m from the bulb.

$$S = \frac{P}{4\pi r^2} = \frac{1000}{4\pi(2)^2}$$

$$S = E \times H = EH \rightarrow (1)$$

$$Z = \left| \frac{E}{H} \right| \rightarrow (2)$$

$$\frac{EH \cdot E}{H} = ? \Rightarrow \frac{1000}{16\pi} \times 377$$

$$E^2 = ?$$

Q. An em radiation has energy 28.8 keV. To which region of em spectrum does it belong.

$$E = \frac{hc}{\lambda}$$

$$28.8 \times 1000 = \frac{6.642 \times 10^{-34} \times 3 \times 10^8}{\lambda}$$

$$\lambda = \frac{6.642 \times 10^{-34} \times 3 \times 10^8 \times 10}{28.8 \times 1000000} = \boxed{0.43 \text{ \AA}}$$

$$= \frac{6642 \times 3 \times 10^{-34+8+1}}{288 \times 10^6}$$

$$\Rightarrow \frac{6642 \times 3 \times 10}{288 \times 10^6}$$

Q. Dist b/w sun and earth  $\rightarrow 1.5 \times 10^{11}$  m. Show that average solar energy incident on earth  $2 \text{ cal/m}^2$ . Calculate energy transfer.

Q. 6 Mega Hz em wave passes from vacuum to a non magnetic medium. If the relative permittivity of air  $= 4$ . Calculate the increment in its  $\lambda$  given -  $\mu_r = 1$ .

Ans  $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$   $\lambda = \frac{c}{\nu}$

$v = \frac{1}{\sqrt{\mu \epsilon}}$   $\lambda = \frac{c}{\nu}$

$\lambda_1 = \frac{1}{\nu_1} \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

$\lambda_2 = \frac{1}{\nu_2} \frac{1}{\sqrt{\mu \epsilon}}$

$\lambda_2 - \lambda_1 = ? \checkmark$

Q. In plane wave, the  $\vec{E}$  oscillates at a frequency of  $3 \times 10^{11} \text{ Hz}$  and amplitude  $50 \text{ V/m}$ . What is the amplitude of  $\vec{B}$ ?